Quick Commerce Sales Analysis Dashboard

Project Documentation

Developed using Microsoft Power BI

Business Intelligence | Data Analytics | KPI Reporting

1. Project Overview

The Quick Commerce Sales Analysis Dashboard is an end-to-end Business Intelligence project developed using Power BI to analyze and monitor the performance of a quick commerce platform. The primary objective is to provide meaningful business insights related to sales, customers, products, stores, and transaction behavior through interactive visualizations and KPI analysis.

1.1 What This Dashboard Helps You Understand

- Overall sales performance across time periods
- Customer purchasing behavior and contribution
- Product contribution to total revenue
- Store-wise sales performance and regional analysis
- Year-over-Year (YOY) growth trends
- Transaction type analysis (Cash, Card, Net Banking)
- Time-based sales patterns by hour, day, and month

1.2 Skills Demonstrated

- ETL Process (Extract, Transform, Load)
- Data Cleaning and Transformation
- Data Modeling and Star Schema Design
- DAX Calculations and Time Intelligence Functions
- Interactive Dashboard Design and KPI Tracking
- Tooltip Reporting and Drill-down Analysis

2. Project Objectives

The following objectives were defined to guide the design and development of this dashboard:

1. Monitor quick commerce sales performance in real time
2. Analyze sales trends across years, months, hours, and locations
3. Compare current year performance with previous year performance
4. Identify high-performing customers and products
5. Analyze transaction patterns such as Cash, Card, and Net Banking
6. Provide interactive filtering and drill-down analysis capabilities
7. Help business users make data-driven decisions efficiently

3. Tools and Technologies Used

The following tools and technologies were used throughout the project lifecycle:

Tool / Technology	Purpose
Microsoft Power BI	Dashboard Development & Visualization
Power Query	Data Cleaning & Transformation
DAX (Data Analysis Expressions)	Measures & Calculations
Excel / CSV Datasets	Source Data
Data Modeling	Relationship Management
Time Intelligence Functions	Year-over-Year (YOY) Analysis

4. Dataset Description

The project uses multiple datasets organized into one Fact Table and multiple Dimension Tables, following a structured data warehouse design pattern.

4.1 Fact Table: fact_table

This is the central table that stores all transactional sales data. Every sale event is recorded as a row in this table.

Column Name	Description
payment_key	Payment identifier — links to transaction details
customer_key	Customer identifier — links to customer info
time_key	Time identifier — links to date/time dimension
item_key	Product identifier — links to product details
store_key	Store identifier — links to store location
quantity	Number of units sold in the transaction
unit	Unit of measure for the product
unit_price	Price per single unit of the product
total_price	Total monetary value of the transaction

4.2 Dimension Tables

Dimension tables provide descriptive context for the transactional data stored in the fact table.

dim_customer — Customer Information

Column	Description
customer_key	Unique identifier for each customer
name	Full name of the customer
contact_no	Customer's contact number
nid	National ID of the customer

dim_item — Product Information

Column	Description
item_key	Unique product identifier
item_name	Name of the product
desc	Product category or description
unit_price	Standard unit price of the product
man_country	Country where the product is manufactured
supplier	Name of the product supplier
unit	Unit of measure for the product

dim_store — Store Location Information

Column	Description
store_key	Unique identifier for each store
state	State where the store is located
district	District where the store is located
area	Specific area or neighborhood of the store

dim_trans — Transaction Details

Column	Description
payment_key	Unique identifier for the payment record
trans_type	Type of transaction (Cash, Card, Net Banking)
bank_name	Name of the bank used for the transaction

dim_time — Date and Time Information

Column	Description

time_key	Unique time identifier
date	Full date of the transaction
hour	Hour of the day when the order was placed
day	Day of the month
week	Week number of the year
month	Month number
quarter	Quarter of the year (Q1 to Q4)
year	Year of the transaction

5. ETL Process

The project followed a complete ETL (Extract, Transform, Load) process before dashboard creation. This ensures data quality and consistency throughout the analysis.

5.1 Extract

Raw datasets were collected and imported into Power BI from structured data sources (Excel/CSV files). All tables were loaded into the Power Query Editor for inspection and transformation.

5.2 Transform

Data transformation and cleaning were performed using Power Query. The following cleaning activities were carried out:

- Corrected incorrect data types on all columns
- Removed null and missing values that could affect analysis
- Removed inconsistent and duplicate records
- Standardized column formats for uniformity
- Cleaned unnecessary whitespace and invalid entries
- Verified key columns used for table relationships

5.3 Load

After transformation, all cleaned datasets were loaded into the Power BI data model. Relationships were then established between tables to enable cross-table analysis and reporting.

6. Data Modeling

After loading the data, relationships between tables were defined in the Power BI Model View. The model is structured for efficient reporting and query performance.

6.1 Schema Type

The data model resembles a Star Schema, but includes an additional Calendar Table between the Time Table and Fact Table for Time Intelligence calculations. This makes it a Hybrid Schema combining both Star and Snowflake patterns.

6.2 Relationships Defined

From Table	To Table	Relationship Type
dim_customer	fact_table	One-to-Many
dim_item	fact_table	One-to-Many
dim_store	fact_table	One-to-Many
dim_trans	fact_table	One-to-Many
dim_time	fact_table	One-to-Many
Calendar_table	dim_time	One-to-Many

7. Calendar Table

A dedicated Calendar Table was created to support Year-over-Year (YOY) analysis and Time Intelligence calculations. Power BI's Time Intelligence functions require a continuous date column with unique date values.

7.1 Calendar Table DAX Formula

```
Calendar_table =
CALENDAR(
MIN('dim_time'[order_date]),
MAX('dim_time'[order_date])
)
```

7.2 Additional Calendar Columns

```
Year = YEAR('Calendar_table'[Date])
Month = FORMAT(Calendar_table[Date], "mmmm")
Month_No = MONTH(Calendar_table[Date])
```

```
Quarter = "Q" & FORMAT(Calendar_table[Date], "Q")
```

8. Calculated Columns

Several calculated columns were added to dimension and fact tables to enrich the data model and support reporting needs.

8.1 dim_time — Date Columns

```
month_name = FORMAT(dim_time[order_time], "mmm")
year = FORMAT(dim_time[order_time], "yyyy")
order_date = FORMAT(dim_time[order_time], "yyyy-mm-dd")
```

8.2 dim_trans — Transaction Display

Combines bank name and transaction type into a single readable label:

```
trans_display = dim_trans[bank_name] & "-" & dim_trans[trans_type]
```

8.3 dim_store — Store Location

Combines district and area into a combined location string for better filtering:

```
store_location = dim_store[district] & " - " & dim_store[area]
```

8.4 fact_table — Price Bucket

This calculated column segments each transaction into a price range category, enabling price-tier analysis:

```
Price Bucket =
SWITCH(
TRUE(),
fact_table[total_price] <= 100, "Very Low",
fact_table[total_price] <= 300, "Low",
fact_table[total_price] <= 500, "Medium",
fact_table[total_price] <= 700, "High",
fact_table[total_price] <= 1000, "Very High",
"Premium"
)
```

Price bucket categories at a glance:

Price Range	Bucket Label
Up to 100	Very Low
101 to 300	Low
301 to 500	Medium
501 to 700	High
701 to 1000	Very High
Above 1000	Premium

9. DAX Measures

DAX (Data Analysis Expressions) measures were created to calculate key business metrics dynamically.

9.1 General KPI Measures

```
TOTAL SALES = SUM(fact_table[total_price])
TOTAL QUANTITY = SUM(fact_table[quantity])
CUSTOMERS = DISTINCTCOUNT(fact_table[customer_key])
Total Stores = DISTINCTCOUNT(fact_table[store_key])
Total Transactions = COUNTROWS(fact_table)
AVG UNIT PRICE = AVERAGE(fact_table[unit_price])
```

9.2 Statistical Measures

```
avg total_price = AVERAGE(fact_table[total_price])
max total_price = MAX(fact_table[total_price])
min total_price = MIN(fact_table[total_price])
```

10. Previous Year (PY) Measures

Time Intelligence functions were used to calculate the same metrics for the previous year, enabling meaningful comparisons. The SAMEPERIODLASTYEAR function shifts the date context back exactly one year.

10.1 Example: PY Sales

```
PY SALES =
```

```
CALCULATE(
[TOTAL SALES],
SAMEPERIODLASTYEAR(Calendar_table[Date])
)
```

10.2 Full List of PY Measures Created

- PY Sales
- PY Quantity
- PY Customers
- PY Average Price
- PY Transactions
- PY Stores

11. Year-over-Year (YOY) KPI Measures

YOY measures compare the current year's performance against the previous year and display the result as a formatted percentage with a visual trend indicator.

11.1 Example: YOY Sales Measure

```
% YOY SALES =
VAR PY = [PY SALES]
VAR A = DIVIDE([TOTAL SALES], PY) - 1
VAR LABEL = FORMAT(A, "#0.0%")
RETURN
IF(
ISBLANK(PY), "NA",
IF(
A = 0, "0.0%",
LABEL &
SWITCH(
TRUE(),
A > 0, " up",
A < 0, " down"
)
)
)
```

11.2 YOY KPIs Included

YOY Measure	What It Shows
YOY Sales	Change in total revenue vs previous year
YOY Quantity	Change in total units sold vs previous year
YOY Customers	Change in unique customer count vs previous year
YOY Average Price	Change in average unit price vs previous year
YOY Orders	Change in total transactions vs previous year
YOY Stores	Change in active store count vs previous year

Each YOY KPI visually signals: Positive Growth (up) or Negative Growth (down) so stakeholders can instantly identify trends without reading numbers.

12. Contribution Measures

Contribution measures calculate the percentage of total sales generated by an individual customer or product, making it easy to identify top performers.

12.1 Customer Contribution

```
Customers Contribution % =
DIVIDE(
SUM(fact_table[total_price]),
CALCULATE([TOTAL SALES], ALL(dim_customer))
)
```

12.2 Product Contribution

```
Product Contribution % =
DIVIDE(
SUM(fact_table[total_price]),
CALCULATE([TOTAL SALES], ALL(dim_item))
)
```

Both measures use the ALL() function to remove filter context so the denominator always reflects the grand total, regardless of slicers applied.

13. Slicers and Interactive Filters

The dashboard provides four interactive slicers that allow users to filter and explore data from multiple angles.

Slicer	Filter Options	Purpose
Year Slicer	Year, Quarter	Analyze performance by time period with hierarchy support
Hour Slicer	Order Hour (0–23)	Understand hourly sales behavior patterns
Transaction Slicer	Cash, Card, Net Banking	Filter analysis by payment method
Store Location Slicer	State, District	Drill-down geographic store analysis

14. KPI Cards

The dashboard header displays the following key business metrics as KPI cards for at-a-glance monitoring:

KPI	Description
Total Sales	Overall revenue generated across all transactions
Total Quantity	Total number of product units sold
Customers	Total count of distinct customers
Avg Unit Price	Average price per product unit sold
Total Orders	Total number of transactions / orders placed
Total Stores	Total number of active stores contributing to sales

15. Visualizations

The following interactive visualizations are included in the dashboard to provide comprehensive analytical coverage:

15.1 Monthly Total Sales Trend

Visualization Type: Line Chart

Purpose: Tracks monthly sales performance over time, revealing seasonal patterns and growth trends

15.2 Customer Contribution to Sales

Visualization Type: Matrix Table

Purpose: Shows how much each customer contributes to overall revenue

Metrics included: Orders, Items, Quantity, Revenue, Contribution %

15.3 Store Location Sales Analysis

Visualization Types: Map Visual + Bar Chart

Purpose: Analyzes sales performance by geographic location

Supports: state-level and district-level breakdowns

15.4 Price Bucket Analysis

Visualization Type: Donut Chart

Purpose: Displays the distribution of sales across price segments (Very Low through Premium)

15.5 Product Contribution Analysis

Visualization Type: Matrix Table

Purpose: Shows which products generate the most revenue and their percentage contribution to total sales

16. Tooltip Features

Two advanced tooltip features enhance the interactivity and analytical depth of the dashboard.

16.1 YOY KPI Tooltip

When a user hovers over any YOY KPI card, a tooltip page appears showing:

- Previous year KPI value for the same metric
- Current year vs previous year side-by-side comparison
- Quick visual identification of growth or decline trends

This was implemented using a dedicated Tooltip Report Page in Power BI.

16.2 Store Location Map Tooltip

When hovering over store bubbles on the map visual, users can see contextual details including:

- State Name
- Total Sales for that location
- Total number of Stores in that area
- Total number of Districts and Areas served

17. Business Insights Generated

Through this dashboard, business stakeholders can achieve the following outcomes:

- Identify top-performing products that drive the most revenue
- Track sales growth trends across months, quarters, and years
- Compare current year vs previous year performance instantly
- Analyze individual customer contributions to total sales
- Monitor payment method behavior (Cash vs Card vs Net Banking)
- Understand regional sales performance across states and districts
- Detect low-performing areas and stores that need attention
- Improve strategic planning and resource allocation through data

18. Key Learnings

This project strengthened the following technical and analytical skills:

Skill Area	What Was Practiced
Power BI Development	Building interactive, production-ready dashboards
Data Modeling	Designing Star/Hybrid schema with proper relationships
DAX Calculations	Writing measures for KPIs, ratios, and time intelligence
Time Intelligence	Using SAMEPERIODLASTYEAR for YOY comparisons
Power Query (ETL)	Cleaning and transforming raw data before analysis
Report Design	Creating clean, consistent, and accessible UI
Business Analytics	Translating data into actionable business insights

19. Conclusion

The Quick Commerce Sales Analysis Dashboard successfully transforms raw transactional data into meaningful, actionable business insights. Through a combination of robust data modeling, carefully crafted DAX measures, and interactive visualizations, the dashboard empowers stakeholders to monitor performance, identify growth opportunities, and make confident data-driven decisions.

The project demonstrates the complete Business Intelligence workflow from source data to final insight delivery:

Phase	Activity

1. Extract	Collected and imported raw datasets into Power BI
2. Clean	Removed nulls, fixed types, and standardized formats in Power Query
3. Transform	Created calculated columns and enriched the data model
4. Model	Built Star/Hybrid schema with proper table relationships
5. Calculate	Developed DAX measures for KPIs, YOY, and contribution metrics
6. Visualize	Designed interactive charts, maps, and matrix tables
7. Deliver	Published an interactive dashboard for business decision-making

This dashboard can serve as a scalable foundation for quick commerce performance monitoring and can be extended with additional data sources, predictive analytics, or real-time streaming data as business needs evolve.